Institute of Information Technology, Noakhali Science and Technology University

Bachelor of Science in Software Engineering

Course Code: SE 4100

Internship Report

Submitted by

Md Masum Billah

ASH2125033M

Submitted to

Internship Placement Office

Institute of Information Technology (IIT)
Noakhali Science and Technology University

Performed at

B-Trac Solutions Ltd

Plot-68, Block-H, Road-11

Banani, Dhaka-1213, Bangladesh

Internship Attended: September 07, 2025 – January 29, 2026

Date of Submission: January 29, 2026

LETTER OF TRANSMITTAL

January 29, 2026

The Chairman
Internship Placement Office
Institute of Information Technology (IIT)
Noakhali Science and Technology University

Subject: **Letter of Transmittal**

Dear Sir,

It is a pleasure to submit the "Internship Report" as per the Internship Program requirement for the course SE 4100 Internship.

This report contains gory details of my activities during the Internship period. I have been working as an intern at B-Trac Solutions Ltd. I was supervised by Md. Mazba Kamal. I hope this report fulfils the requirements of the Internship Program.

I am requesting that you accept and consider this report.

Yours sincerely

Md Masum Billah
ASH2125033M
Session: 2020-21
BSc. in Software Engineering Program
Institute of Information Technology (IIT)
Noakhali Science and Technology University

DECLARATION

TO WHOM IT MAY CONCERN

I, **Md Masum Billah**, bearing ID: ASH2125033M, BSc. in Software Engineering Program, hereby declare that the presented "Internship Report" is uniquely prepared by me after completion of six months' work in **B-Trac Solutions Ltd**.

My original work is submitted to the Institute of Information Technology (IIT), Noakhali Science and Technology University, and no part of the report has been submitted for any other degree or fellowship, and the work has not been published in any journal or magazine.

Md Masum Billah
ASH2125033M
Session: 2020-21
BSc. in Software Engineering Program
Institute of Information Technology (IIT)
Noakhali Science and Technology University

LETTER OF ENDORSEMENT

TO WHOM IT MAY CONCERN

This is to certify that **Md Masum Billah** was an intern at **B-Trac Solutions Ltd**. During this period, I was one of his supervisors.

I hereby confirm that I have reviewed the entire report. The contents of this report are true and not confidential to the company. The projects and training assignments mentioned in this report had the successful participation of Md Masum Billah.

I wish him all the very best for his future life.

Md. Mazba Kamal
Deputy Manager
B-Trac Solutions Ltd

Safayet Abdullah
Head of Projects
B-Trac Solutions Ltd

Certificate of Approval

This Internship report, submitted by **Md Masum Billah, ID No: ASH2125033M,** to the Chairman of the Internship Placement Office, Institute of Information Technology (IIT), Noakhali Science and Technology University, has been accepted as satisfactory for the partial fulfilment of the requirements for the degree of Bachelor of Science in Software Engineering and approved as to its style and contents. The presentation was held in January 2026.

Internship Examination Committee:

Mohammed Nizam Uddin, Chairman

Associate Professor & Director (Acting)
Institute of Information Technology (IIT)
Noakhali Science and Technology University

Md. Iftekharul Alam Efat, Committee Member

Assistant Professor
Institute of Information Technology (IIT)
Noakhali Science and Technology University

Md. Eusha Kadir, Committee Member

Assistant Professor
Institute of Information Technology (IIT)
Noakhali Science and Technology University

Tasnim Rahman, Committee Member

Assistant Professor
Institute of Information Technology (IIT)
Noakhali Science and Technology University

DEDICATION

I dedicate this internship report to my parents, whose unconditional love, sacrifice and constant encouragement, have made me what I am today as an academic and a person. They always encourage and trust me, which has made me study diligently and rely on myself to achieve my education.

This is one of the reflections of their guidance and patience, which have influenced me to survive the difficulties and aim at excellence. I owe a lot to them as well, because they have helped me to develop my values, discipline and commitment to ensuring that I accomplish my professional objectives.

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OFFER LETTER

Date: September 24, 2025

To
Director,
Career and Placement Center (CPC),
Noakhali Science and Technology University (NSTU)
Bangladesh.

Subject: <u>Confirming letter of internship.</u>

Dear Sir / Madam,

We would like to inform you that Md. Masum Billah, ID no: ASH2125033M, Session: 7th Semester, contact no: 01400095352 has started his internship program with B-Trac Solutions Ltd. in the Web Applications department. The duration of the internship program is six months.

Thank you,

Md. Tangir Hossen
Strategic HR Partner

B-Trac Solutions Ltd.
Plot-68, Block-H, Road-11, Banani, Dhaka-1213, Bangladesh
+8802-883 8001-4 +8802-883 6868 info@btracsl.com
www.btracsolutions.com

ACKNOWLEDGEMENT

I would like to thank B-Trac Solutions Ltd. for providing me with the opportunity of doing my internship program under a professional and supportive environment. The organisation significantly provided me with an excellent opportunity to apply my theoretical knowledge to actual projects and this brought me a good exposure to the industry and relevant experience on software engineering.

I particularly like to thank Md. Mazba Kamal (Deputy Manager), my mentor during the internship. His consistency in directions, technical suggestions, and positive criticism were indeed a motivator to my learning. He not only made me more skilled in technical aspects, but also showed me how to cope with professional responsibilities and fulfill the standards of the industry.

I would also like to thank Md. Iftekharul Alam Efat, Assistant Professor at the Institute of Information Technology (IIT) of Noakhali Science and Technology University (NSTU) with great appreciation as well as academic supervision and encouragement. His assistance bridged the gap between my classroom experience and experience in the real world and it was what made this report complete.

PREFACE

One of the reasons why this internship report was written is to meet the academic specifications of my undergraduate program. The internship was an extremely important part of my academic life, as it allowed me to test the knowledge that I had received in the coursework in practice. I have documented my experiences, duties, learning outcomes, and general development as a professional during the internship process.

I completed my internship at B-trac solutions Ltd as an intern Software engineer. My primary aim was to have first hand experience with the software development industry and to observe how our theories are applied in real-life situations. Having participated in actual projects, I found out how the industry processes operate, how development procedures are implemented, and what the professional standards are like in a technology driven firm.

The internship provided me with some development activities that contributed to the actual value of projects of the company. These projects provided me with invaluable experience on software engineering concepts including design, code, deployment and collaboration in a multidisciplinary team. The other difficulties I experienced included learning new technologies, understanding difficult systems, and adhering to professional expectations, which were necessary components of my learning process as well.

EXECUTIVE SUMMARY

This internship report provides an in-depth account of my internship experience as a Software Engineer Intern at B-trac Solutions Ltd. My undergraduate studies have the internship as part of their requirements. I describe the company profile, what I was doing, my duties, the learning outcomes and the professional development that I attained over this time.

To close the gap between the academic study and the actual practice was the main goal of the internship. All the time, I had to work on projects, which presupposed using web technologies, API integration, and the best practices of software engineering. Another professional area that I concentrated on was teamwork, communication, time management, and problem-solving.

In achieving these objectives, I did things in a hands-on, experiential way. I was an active project developer with the supervision of an experienced mentor. The practical work, frequent discussions, review of the code, and teamwork were all the sources of learning. I further gathered additional data through company documents, official books and discussion with colleagues and other interns.

The predominant outcomes of the internship would be the enhanced technical expertise of modern software development tools, clearer insight into the processes of working in the industry, and increased self-confidence in the ability to cope with the professional responsibilities. The experience with producing the level of production and real-time systems enhanced my problem-solving and analytical skills.

Generally, the B-Trac Solutions Ltd internship was an important learning experience that has greatly contributed to the development of my technical and professional abilities. The experience and capabilities that I have acquired are putting me on my feet to do academic projects in the future and to work in the field of software engineering. I conclude that industry based internships are paramount in ensuring that one comes up with graduates that are not only competent but also prepared to work.

Introduction

Preamble

Since the start of my first semester, I have been working on the groundwork of software engineering. Initially, the coursework was completely theoretical, including structured programming, data structures, algorithms, object-oriented programming, databases and design patterns. Throughout the semesters, I was able to observe how those concepts were being actualized in the form of labs, assignments, and project work that slowly transformed abstract concepts to real practical skills.

Throughout the six semesters, faculty, a well-structured curriculum and frequent examinations challenged me into exhibiting more analytical skills, problem-solving abilities and an overall comprehension of what it entails being an engineer. However, despite all that theoretical training and those projects I have done, I understood that the real software development in the industry possesses levels of complexity that cannot be fully reflected in a classroom.

The last and the most important phase of this learning struggle is the academic internship. It helps in relating university education to life practice since it allows me to put what I have learned into real practice in the industry. This experience has played a central role in getting ready for my future career, exposing me to real issues, duties as a professional and the teamwork work ethic, which characterizes the profession. By undertaking the internship as a component of my coursework, this will not only fulfill the theoretical base but also enhance job preparedness since whatever I study and what the employer requires are in line.

Origin of the report

This report is done under a compulsory requirement in my undergraduate course internship. It captures the experience that I had during my internship, the organization that I had internship experience in, the assignments that I was assigned, the learning outcomes, and the professional skills that I gained during the internship. It is an academic documentation whereby the internship assisted me to form my personal, technical, and professional development, meeting the faculty reporting time.

Objective of the internship

The internship was intended to provide practical exposure to the world of software, improve my technical skills and develop professional skills including teamwork, communication, and responsibility. The other important objective was to acquire the ethics of the workplace, organizational structures, and knowledge on how to solve real life problems, which is necessary to facilitate the transition between the life of a student and the work environment.

Scope

It was not only a polishing of technical skills during the internship. It enabled me to bridge the gap between what I learned in classrooms and how it can be applied to actual problems, work on a team, and understand how theory can be applied in real-life software development. I have gone through industry standard workflow, tools and development practices, and I have developed a network that can assist in future employment opportunities.

Methodology

I used qualitative experience in forming this report. My daily job, discussions with mentors and team meetings are the main sources of the core information. I also obtained background information about the organization through the organization site, LinkedIn and Facebook in order to learn more about the organization in terms of culture and practices. This combination of my personal experience, peer-to-peer communication, mentorship, and authorized materials is the foundation of what I represent here.

Limitations

Although the internship provided a good learning experience, there were some limitations to my internship job. I primarily addressed set tasks and overseen projects, without being engaged in core operations or higher-level decision making. Therefore, I was not present at all internal workflows and strategic arguments, as well as significant project planning.

Also, data-security and confidentiality regulations limited my access to sensitive data, proprietary systems and major business information. As such, this report merely talks about the areas and responsibilities that were officially a part of my internship experience.

Recruitment

Overview

The internship is the result of the good partnership between our institute and the leading software companies. These ties had already landed internships and jobs through past classes and consequently, the industry trusts us, and we trust them. Due to that, companies would like us to accept internship opportunities as a part of our course work keeping the program in accordance with our studies and employer wants.

On-Site Interview

In the recruitment, our faculty directed our resumes to a group of our partners such as Brain Station 23, Samsung R&D, Brac IT Services and other technological firms. The initial screening was followed by those that passed being called in to take the company specific tests and interviews all within the respective procedures of the company.

The tests typically contained technical questions of Q&A, coding, and behavioral questions. Conducting such interviews provided us with an idea of the actual hiring requirements and enabled us to strengthen the relationship between our institute and the companies. It also enhances our career preparedness as we are informed of what the employers require even before we complete our studies.

On-Campus Recruitment

In the case of the on-campus segment, 9 companies initially expressed interest, but in the end, only 3 firms, namely, B-Trac Solutions Ltd, OS IT and Acquant Technologies, showed up on campus to recruit. It was a huge win to us, as it increased the popularity of the institute and offered us more employment opportunities.

OS IT has the vast majority of its internships remote, and they came to campus on a short interview. The selected students of our faculty were interviewed with CEO, Mohammad Moniruzzaman, and the Director, Nusrat Jahan.

Acquant Technologies conducted an interview and then a task. The chosen ones from the interview were given a development task and the ultimate grades were based on the way they completed the task. The participants were the CTO, Mahabub Alam Tito, COO, Mahafuzur Rahman, and Web and Digital Strategist, Robiul Awal.

The path that B-Trac Solutions Ltd used is another one since it concentrated on creativity, communication, and mindset rather than on mere technical tests. They established a group chat rather than regular interviews. The head, whom we interviewed, was the CEO Mohammed Tanvir Siddique, who talked us through our studies, how we think, what makes us work, and where we see our career going.

Personal Experience

I was able to sit in some interviews throughout the entire recruitment process. The interrogatives at Samsung R&D were largely probing my resume, the work I had completed, and my Laravel skills - as that is what the role involves. They also struck me on my clubs and competitive coding and I realized that having a well-balanced profile is very important.

Webase Technologies was also one of the companies that received my CV, which was the first online 10-minute chat. The questions they posed were primarily behavioral and they test my knowledge of Laravel since that is what they need.

The on-campus recruitment also enabled me to discuss with Acquant Technologies. A talk of that was principally a behavioral one, succeeded by an actual trial.

The best experience was the group discussion with the CEO of B-Trac Solutions Ltd. We talked about all our course work and practical skills to how we could assist the company, our career goals, what they would expect of interns, and what was the next project. It was a true eye opener into the line of thinking the business has as well as what they are targeting.

Final Selection and Appointment

Following the on-campus and off-campus recruiting, I was able to secure a Software engineer intern position at B-Trac Solutions Ltd. I received a call about the selection by a HR representative Afsana Akther, who informed me that I had been selected and sent me the details of the start.

Parent Company

Overview of Bangla Trac Group

Bangla Trac Group is a top and diversified group of Bangladesh, and it has existed for eighteen years, which is quite an impressive effect in the national economy and a number of major industry areas.

It began in 2007 when Bangla Trac Communications was licensed as an International Gateway (IGW) service provider and has since that point been a multi-sector operation with interests in technological, energy, infrastructure and retail.

The contribution made by the group to the country is massive particularly in power generation, telecom, IT, and infrastructure. They contribute to the national development through a provision of trustworthy services in the most important spheres driving the development of the economy and technological advances. They have been concerned with quality, innovation, and customer oriented services and this has contributed to the industrial power and services delivery in Bangladesh.

At the moment, many business units of Bangla Trac Group deal with power generation, IT and digital solutions, telecom, and international retail franchises. The strategic vision of them is focused on the sustainable development, technological integration, and the creation of global brands on the national market. They are continuously broadening their base as long as it does not compromise their economic growth and service quality.

Business Area

The business of Bangla Trac Group is distributed in various large sectors that contribute to the construction of the infrastructure in the country and services innovations.

- **Power and Infrastructure:** The power units will contribute to the national grid a gigawatt of power that will accommodate the supply of power and infrastructure improvements.

- **Information Technology and Digital Services:** The organization provides digital solutions, software/hardware choices, and IT solutions to both government projects and the private business.

- **Telecommunications and Connectivity:** Their telecom projects provide international gateway services, data hosting and network solutions which enhance connectivity inside and outside Bangladesh.

- **Retail and Franchise Services:** It operates international retail franchises, big global brands among others contributing to the local economy in terms of service, employment, and consumer interaction.

All these sectors are aligned to the overall objective of the group, which is to play a part in the socio economic development of Bangladesh by an amalgamation of various but combined business operations.

Sister Company

Bangla Trac Group also has a few affiliated and subsidiary companies with each having different industries and contributing to the diversified portfolio of the group.

Figure 01: Sister Company of Bangla Trac Group

B-Trac solutions Ltd: It is a technology services company dealing in software development, digital solutions and IT consulting.

B-Trac Technologies Ltd: This is an IT distribution firm with a value added business that coordinates contracts with major brands in the world.

Bangla Trac Communications Ltd: A pioneer telecom and international gateway service provider.

Bangla Trac Engineering Ltd: Bangla Trac Engineering ltd is a company dealing with engineering solutions, manufacturing as well as support services to industrial and power systems.

These companies combined expanded the membership of the group in the tech, power, engineering and service sectors strengthening its position as a significant corporate player in Bangladesh.

Partner Company

Bangla Trac Group collaborates with other international and domestic entities to increase its coverage and service capacity.

Figure 02: Partnar Company of Bangla Trac Group

Leading Technology Vendors: The group partners with the world renowned brands of Dell, HP, Microsoft, Panasonic, AVITA, and BlackBox and is supported by their affiliate companies such as B-Trac Technologies Ltd, to transform digital and distribute IT in Bangladesh.

Telecommunications and Data Service Entities: The relationships with the telecom and international gateway collaborations assistance enable the group to offer substantial network connectivity and data solutions.

Such alliances help the group to provide comprehensive solutions and maintain high standards of service in various segments.

Company Profile

Overview of B Trac Solutions

B-Trac Solutions Ltd is a technological company based in Dhaka, Bangladesh which specializes in the design, development and deployment of software and application systems in web, mobile and the IoT. It was originally an IT centre of Bangla Trac Group and has been driving the digital tidal wave with providing new innovative software to the governments, corporate organizations and other large companies.

The firm has actually left a mark in the Bangladesh IT sector by influencing people to embrace locally developed digital systems which address actual business and social issues. Their solutions allow keeping the user in mind and prioritize the rock solid cloud integration and flexible technology in the frontline, and therefore allow decision-makers to utilize data to get ahead and streamline the operations of the public and the private sector.

B-Trac Solutions is yet another company that enhances the bigger Bangla Trac Group with the needed tech and digital services punch to the diversified portfolio. The resulting synergy enables the group to distribute end to end solutions to combine software innovation with the existing infrastructure, telecom, and enterprise services in supporting growth of the national economy and enhancing industry standards.

Mission and Vision

Vision: To be a vibrant global technology firm that facilitates stakeholder prosperity with excellence, integrity, humility and people centric leadership.

Mission: To transform and be an add value to the user experience through addressing real life challenges through technology and innovation.

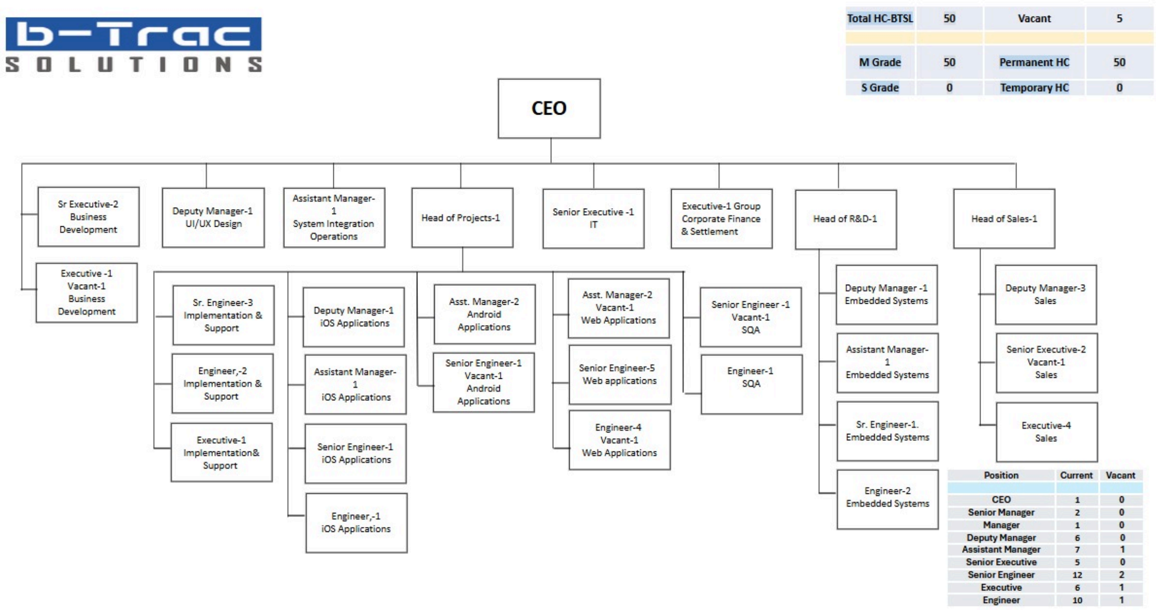
Strategic Focus and Core Values:

B-Trac Solutions Ltd operates under a number of principles that govern its developmental processes and relationships with clients:

- **Innovation:** We are continuous with innovation that drives us to provide game changers.
- **Efficiency:** The automation and workflow optimization are the key factors to improved performance.
- **Reliability:** It is a good pledge to provide reliable and smooth solutions.
- **Customer Centricity:** Concentrate on the needs and surpass the customer expectations.
- **Collaboration:** Strategic partnerships and teamwork enable sustainable growth and long-term impact.

Company Hierarchy

The structure of the organization is established to facilitate strong leadership, responsibility and the best technical ability. Mohammed Tanvir Siddique, the CEO, is the leading boss, and the projects, R & D, sales, IT, and corporate finance senior managers are the next bosses.



Total HC-BTSL	50	Vacant	5
M Grade	50	Permanent HC	50
S Grade	0	Temporary HC	0

Position	Current	Vacant
CEO	1	0
Senior Manager	2	0
Manager	1	0
Deputy Manager	6	0
Assistant Manager	7	1
Senior Executive	5	0
Senior Engineer	12	2
Executive	6	1
Engineer	10	1

Figure 03: Hierarchy of B-Trac Solutions Ltd

Under these leadership roles, the organization is divided into specialized technical teams such as Web Applications, Android Applications, iOS Applications, Embedded Systems, Software Quality Assurance (SQA), System Integration, and Support Services.

Services

B-Trac solutions Ltd implements a robust set of technology solutions to provide end to end digital solutions:

- Tailor made Web Application Development.
- Android and iOS Mobile App Development.
- System Integration solutions.
- Infrastructure and Cloud Services support.
- IT Advisory and digital transformation.
- Technology implementation and support services.

Such services assist businesses in modernising, increasing their output, and getting on scalable digital platforms.

Products

The company has created and introduced some digital goods that solve issues in the real world:

Figure 04: Products of B-Trac Solutions Ltd

- **Carpoplo:** Vehicle safety and tracking solution.
- **Yes Parking:** Smart parking management system.
- **Nirapath:** Personal safety and emergency response app.

- **Workopolo:** Task and labour management.
- **Seemo:** Visual security and monitoring solution.

All products reflect the bite of the company in terms of usability, reliability and technological innovation.

Projects

B-Trac Solutions Ltd has been providing numerous projects in both web and mobile. The focus of the projects is on:

Figure 05: Projects of B-Trac Solutions Ltd

- **CIMS:** Citizen Information Management Web-Based.
- **DMP Duty:** Duty tracking and task tracking App.
- **Banani Club:** Web based and mobile application.
- **Doorstep:** Web based service delivery application.

The projects demonstrate how the company will be able to establish scalable, secure and user facing systems.

Partner Company

The B-Trac Solutions Ltd collaborates with the leading global technology suppliers to provide high quality IT, system integration, and cyber-security solutions. Large allies are Dell, HP, Barracuda, Red Hat, Teknowledge, Sirius International Holding and Presight.

Figure 06: Partner Company of B-Trac Solutions Ltd.

These partnerships have enabled B-Trac to distribute enterprise-level solutions that align with the changing requirements of government and corporate customers.

Awards and Participations

B-Trac Solutions Ltd has leaped into local and global technology events, and has won accolades on both innovation and excellence:

Figure 07: Awards of B-Trac Solutions Ltd.

- **National Hackathon** - 2016 (Award)

- **PTC Innovation** - 2019 (Participant)
- **Chittagong Motor Fest** - 2022 (Participant)
- **ITEX** - 2023 (Award)
- **Chittagong Motor Fest** - 2024 (Participant)
- **Dhaka Motor Fest** - 2025 (Participant)

These victories support the fact that the company strives to improve and become a tech leader.

Internship at B-Trac Solutions LTD

Overview

The B-Trac Solutions Ltd internship was closely related to my course work at the university in software engineering. Some of the core topics covered like structured programming, web technologies database management system, software requirements specification (SRS), and software testing and quality assurance were directly used in my assignments. This on-the-job training helped to fill the gaps in my theoretical knowledge acquired during classes and develop my interest in problem-solving, using real projects. This means that the internship was a positive experience of bridging the gap in academic knowledge and professional practice and enhancing my technical ability and conceptual coherence.

Internship Duration

This internship was five months, between 7 September and 29 December, as per the requirements of the university. The organization also provided placements in a number of technical divisions, such as Embedded Systems, Android Development, Web Development and other specialty software related fields, which enabled the interns to gain a comprehensive perspective on the various technical activities of the organization.

Joining Day

The first thing on my first day was to report to the office at 10:00 A.M. When I arrived, I received an introduction from the HR, Afsana Akther, who took me through the administrative processes required of me and formally introduced me to the company policies, rules of work, and expectations of the profession. This made me have a clear idea of the standards and behavior expected of all the employees in an organization.

Upon undergoing the administrative formalities, I was sent to the Md. Mazba Kamal (Deputy Manager) who introduced me to the Development Team as well as to the structure of that team and my position as an intern in the team being a Software Engineer. This orientation has made me feel welcomed and ready to start my internship professionally and in an organizational way.

Initial Guidance

In the first stage, I was officially instructed by Safayet Abdullah (Head of Projects) regarding professional behavior, performance of tasks and duties in the workplace. His straightforward guidelines on how organizations work, on project procedures, and expectations of the interns have made me realize my position within the team and adjust to a work environment

Afterward, Md.Mazba Kamal looked at my resume and evaluated my technical experience. On the basis of this assessment, I was given a full-stack development position and listed my mentor as official.

Roles and responsibilities

WorkingThe assignment in the Web Development Team gave me an exposure to the state of the art of web technologies, frameworks of web development, and group practice of software engineering, which added more weight to my hands-on skills and resonated well with the university curriculum.

The tasks I was responsible for as a Software engineer Intern were:

- Development and design of full stack web applications, as per project requirements.
- Producing quality code of acceptable standards that is easily readable and maintainable.
- Checking and testing applications to provide functionality, performance and reliability.
- Fostering testing efforts in order to stabilize and ensure quality of applications.
- Implementing applications in accordance with the standard deployment procedures.
- Preparation of development procedures and proper project documentation.

The intern was supposed to conform to established working processes, code standards, and good record keeping. With time, my role would shift to include independent problem-solving and proactive participation in project modules that helped me to improve my technical expertise and professional responsibility.

Personal Growth

The internship with B-Trac Solutions Ltd has helped me to extensively grow both personally and professionally. The exposure to real world software development issues sharpened my problem solving skills, analytical skills, and flexibility. The experience of working in a professional development setting helped me learn more about the expectations in the industry and enhance my task management skills.

The staff applied the principles of agile development, and it enabled me to observe and be involved in the development cycles, sprint planning, and group work. The modern technologies that I tried and worked with throughout the internship included GraphQL APIs, GitHub GraphQL Public API, and Firebase Cloud Messaging (FCM). These experiences have expanded my technical expertise as well as increased my confidence in the use of modern development tools and architectures.

Work Environment

B-Trac Solutions Ltd has an adult, cooperative, and supportive working environment. The members of the team are friendly and they do not hesitate to share knowledge and work as a team. Constructive feedback and technical guidance are given to the supervisors on a regular basis and this encourages ongoing learning and improvement.

It has a structured and flexible culture, and the normal working hours are between 9:00 A.M. and 6:00 P.M. The work environment is more of a co-working environment where there are no assigned desks and everyone works together and communicates well. The organization takes weekends on Friday and Saturday, and every week stipulates one day of remote work (home office). This level of balance has a positive impact on productivity, work-life balance, and overall satisfaction.

Facilities

B-Trac Solutions Ltd has sufficient facilities that can guarantee a comfortable and productive working environment. There is indefinite coffee and lunch offered in the office, and lunch is normally served after 2:00 P. M. The work environment will be structured to facilitate cooperation, teamwork and communication among the employees which will bring a good culture and increased efficiency.

Farewell

A formal farewell ceremony was conducted at the end of the internship period, which reflected on my performance, learning outcomes and improvement areas that I need to strive on. It was also a positive and encouraging end to the professional experience as the farewell made the organization realize my commitment to their work and contributions during the internship period.

Conclusion

The internship with B-Trac Solutions Ltd was an enriching experience that was valuable and which succeeded in integrating the academic knowledge with the field exposure. The internship was essential in the progression of my technical skills and professional maturity through the structured training and meaningful responsibilities, as well as a supportive environment, which can be a great foundation in the future of my academic study and a career in software engineering.

Practice Projects

Taskify

Taskify is a command line application that I created to manage Firebase projects. It allows me to create, view, and delete Firebase projects with the help of service-account JSON files. I included validation checks to ensure credential is right and I save project data in a persistent manner to ensure that I can easily manage them in the long run.

Motivation

As part of my internship responsibilities, I was assigned to work on an earthquake alert system intended to notify various mobile applications in real time. To make this possible, I was required to maintain records of Firebase credentials of multiple apps. The experience of that prompted me to write a tool that would be able to handle Firebase service accounts either through a GUI or through the command line using *commander.js*. I chose to develop a CLI because it is unfamiliar to me and presented an opportunity to experiment with another paradigm of developing other than the typical UI based workflows.

Features

The key features of Taskify include:

- Interactive REPL mode (terminal stays alive)
- Register Firebase projects using service account JSON
- Validate Firebase service account structure
- List all registered projects
- View details of a specific project
- Remove individual or all projects
- Persistent local storage (*projects.json*)
- Type-safe implementation with TypeScript
- Simple, predictable CLI interface

Technologies

- Node.js

- Commander.js
- TypeScript

Learnings

During the creation of this application I acquired practical knowledge of the construction of command-line applications with Node.js and TypeScript. I also learned how to design a centralized logging system using *Winston*, implement input validation, and manage persistent local data storage. The project enhanced my knowledge on the topic of backend tooling, automation and effective management of project configuration within the real-world software systems.

Proof of Work

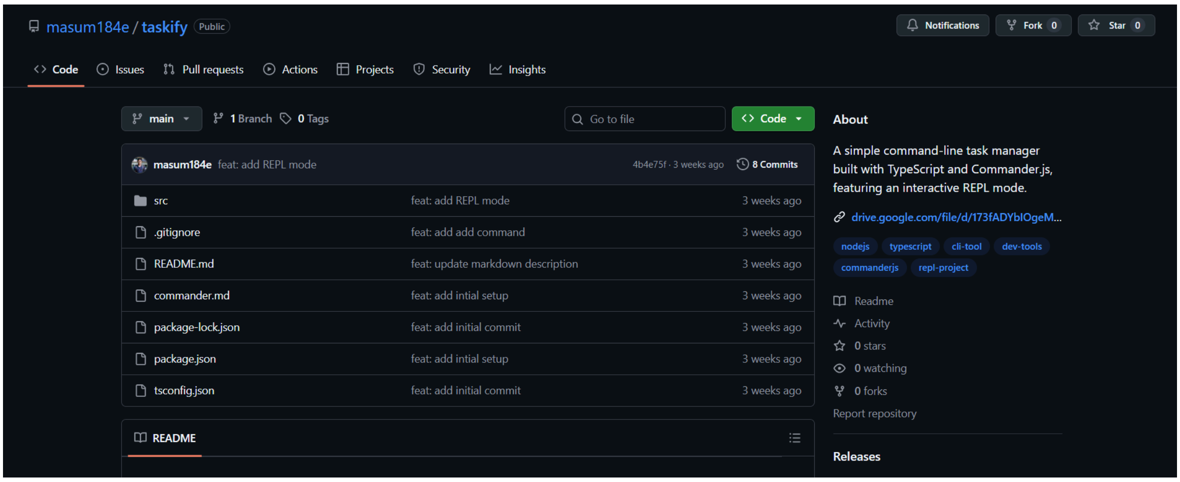


Figure 08: GitHub repository of Taskify

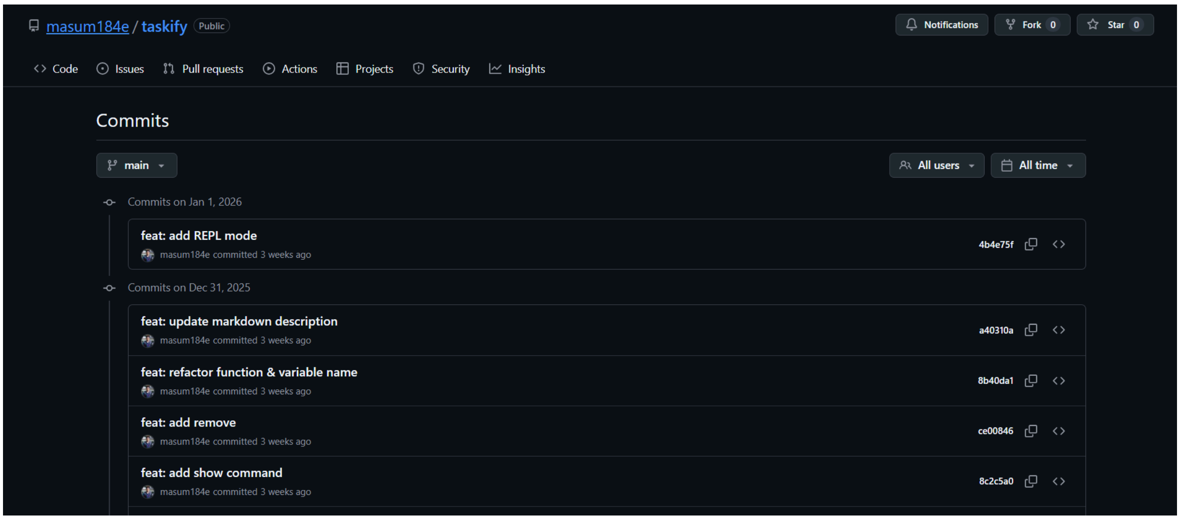


Figure 09: Commit history of Taskify

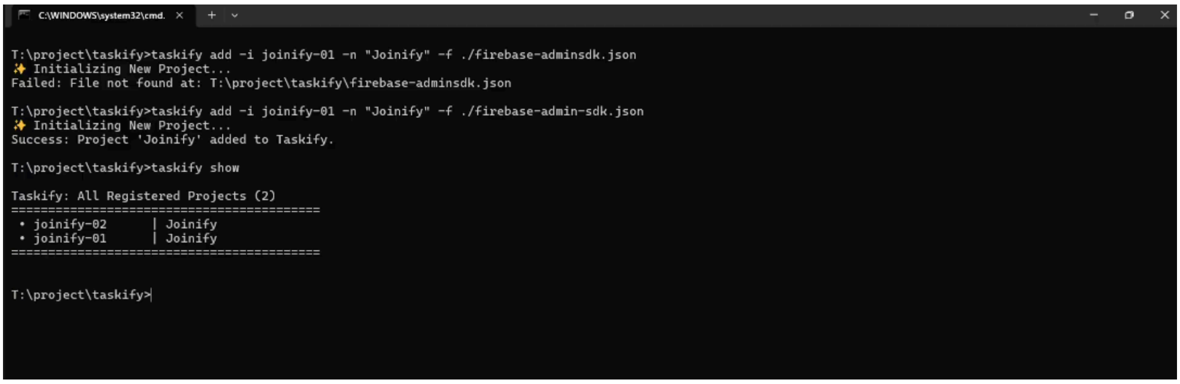


Figure 10: Command line interface of Taskify

GitHub Repository: https://github.com/masum184e/taskify

Video Preview: https://drive.google.com/file/d/173fADYbIOgeML2ZwCgnRUZ_rQLnJq4xk

FCM Playground

FCM Playground is a sandbox application that I developed as an experiment of Firebase Cloud Messaging (FCM) and how to use the Firebase Client SDK and Firebase Admin SDK. It illustrates the entire process of client-device registration, foreground and background notification processing and messages relayed by a server side context. This was aimed at trying to replicate real life notification delivery conditions in a controlled laboratory experiment.

Motivation

During the development of the earthquake alert system, I had to have a sound method of testing the process of sending and receiving alert messages through FCM. I have found no existing tools that would allow me to receive FCM notification on the devices and send messages using a web interface since none of them fulfilled my needs. Then I chose to create my own testing platform and this provided me with a more technical understanding of FCM and a useful tool that can be reused by the alert system.

Features

Client SDK

- Intelligent entry and authorization of customer credentials.
- Receive messages through Firebase Client SDK.
- Recall and reveal FCM device token.
- Subscribe to a messaging topic(s)

Admin SDK

- Validation and configuration of the Firebase Admin SDK
- Send messages to a specific device using its FCM token
- Send messages to a group of devices based on topic subscription
- Broadcast messages to all registered devices, with a default subscription topic (*all_users*)

Technologies

- Next.js
- Firebase
- Shadcn UI
- Zod

Learning

Developing this app I managed to acquire a certain amount of practice in applying Firebase Cloud Messaging and dealing with real-time communication between servers and client devices. I have formed a clear understanding of the way FCM manages foreground and background notifications and how the *firebase-messaging-sw.js* service worker contributes to the messaging of web based applications. I also got to know how browser APIs such as Navigator interface can work with service workers in order to purpose real time notifications. This project enhanced my understanding of real time messaging systems and event driven systems of communication architecture to a significant extent.

Proof of Work

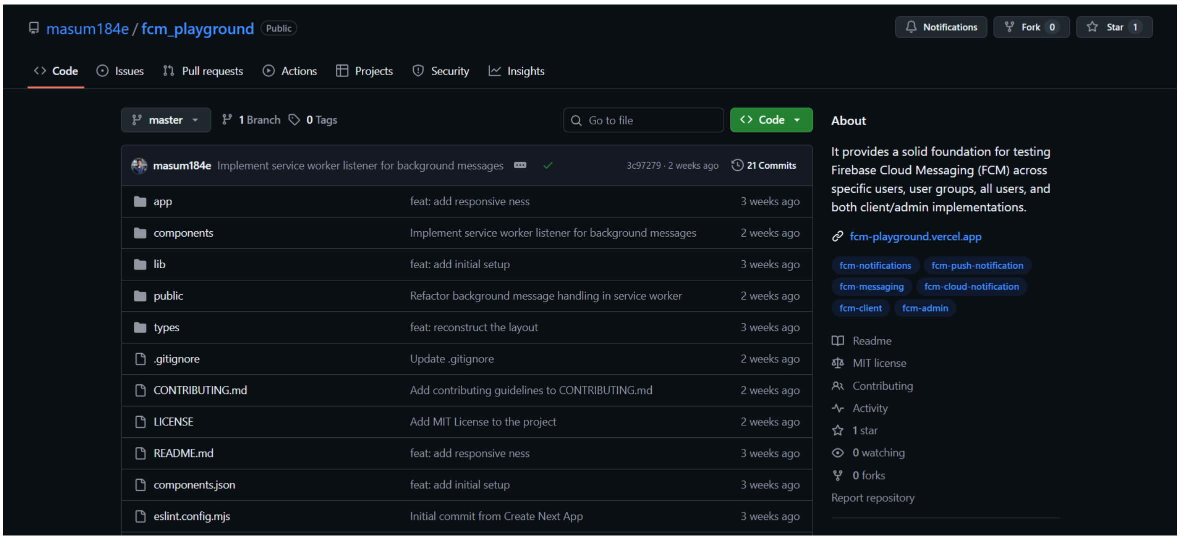


Figure 11: GitHub repository of FCM Playground

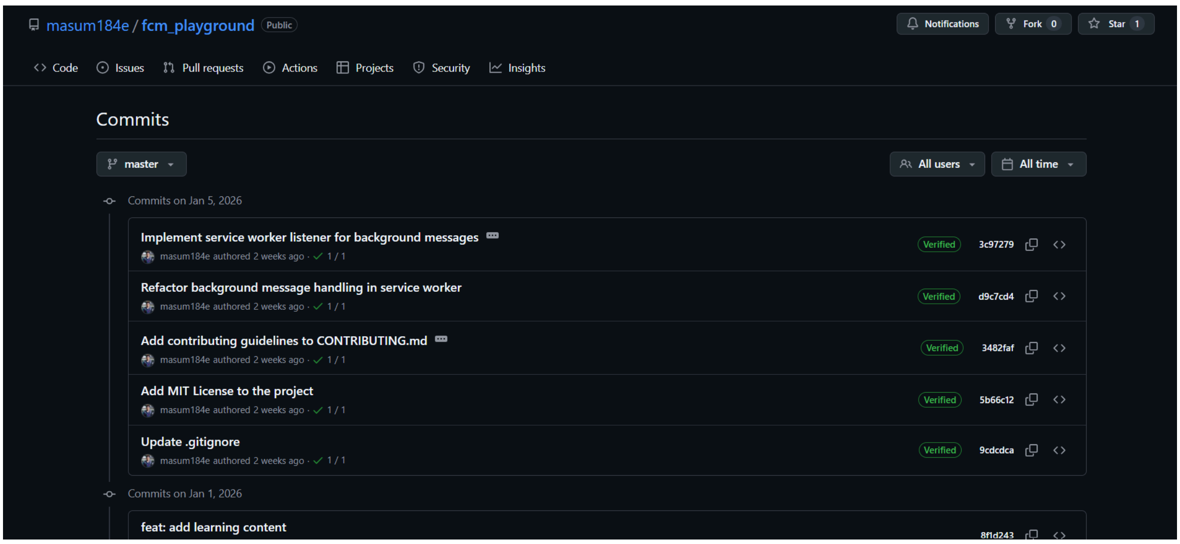


Figure 12: Commit history of FCM Playground

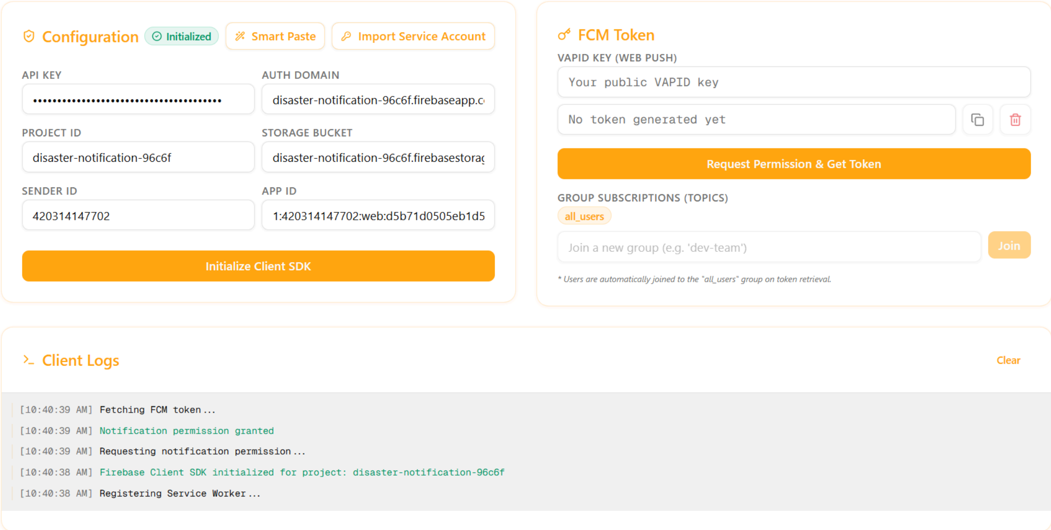


Figure 13: Client SDK interface for FCM Playground

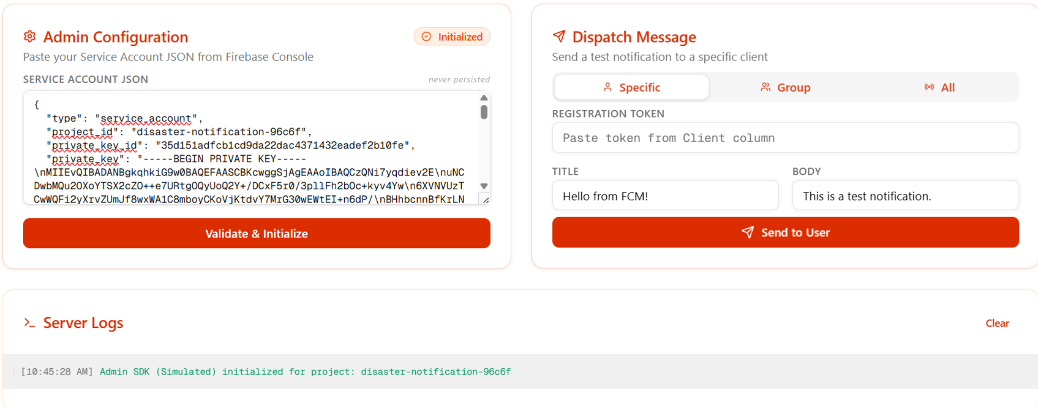


Figure 14: AdminSDK interface for FCM Playground

GitHub Repository: https://github.com/masum184e/fcm_playground

Live URL: https://fcm-playground.vercel.app/

Assigned Projects

BTSL GitHub Dashboard

The BTSL GitHub Dashboard was a web based tool of analysis that I constructed throughout the internship period in B-Trac Solutions Ltd. The entire idea was to provide team leaders with a means to check on the activities of all members of the GitHub organization so that they could be able to coordinate and monitor projects and assess performance in a simpler manner.

Features

The key features of the BTSL GitHub Dashboard include:

- Track recent activities in repositories, including commits, pull requests, and issues
- Analyze PR trends by status (open, closed, merged), labels, and selected time periods
- Measure issue activity (open vs closed) and label based categories
- Monitor code contribution trends using commit history, additions, and deletions
- Identify recent active repositories based on the latest push dates
- Profile teammates with detailed contribution metrics across commits, issues, and PRs
- Evaluate team collaboration through PR reviews and review timelines
- Granular filtering by repository, contributor, label, and activity type for deeper analytical insights

Technologies

- Next.js
- GraphQL (GitHub Public GraphQL API)
- Docker
- Shadcn UI
- TypeScript

Learning

I had previously been exposed to GraphQL on full-stack applications (where I was allowed to experiment with the API and queries). I was now forced to deal with the public GitHub API, and this came as an entirely new ball game.

At first I was having difficulties writing queries that would satisfy the exact requirements, but after browsing the Apollo GraphQL API Explorer I began to construct decent queries more quickly. That resource actually boosted my mastery of using public GraphQL services, not only GitHub, but other APIs as well.

Another lesson that I gained during the development of the backend logic is how the GitHub API can be paginated. I needed to work with massive data pulls, and this furthermore enhanced my knowledge of fetch strategies and performance optimization.

Then, I used company credentials to deploy the app to a VPS after development. It worked well on the local machine but when I switched to the VPS the UI would fail to load styles due to a proxy issue; I traced this issue and corrected it.

Subsequently, I encountered inconsistencies in API behavior, where some requests succeeded while others failed. Through detailed debugging, I determined that the issue stemmed from differences between Next.js client-side and server-side components, specifically related to network request handling. Resolving these issues enhanced my understanding of Next.js client and server components, deployment environments, and real-world production debugging.

Proof of Work

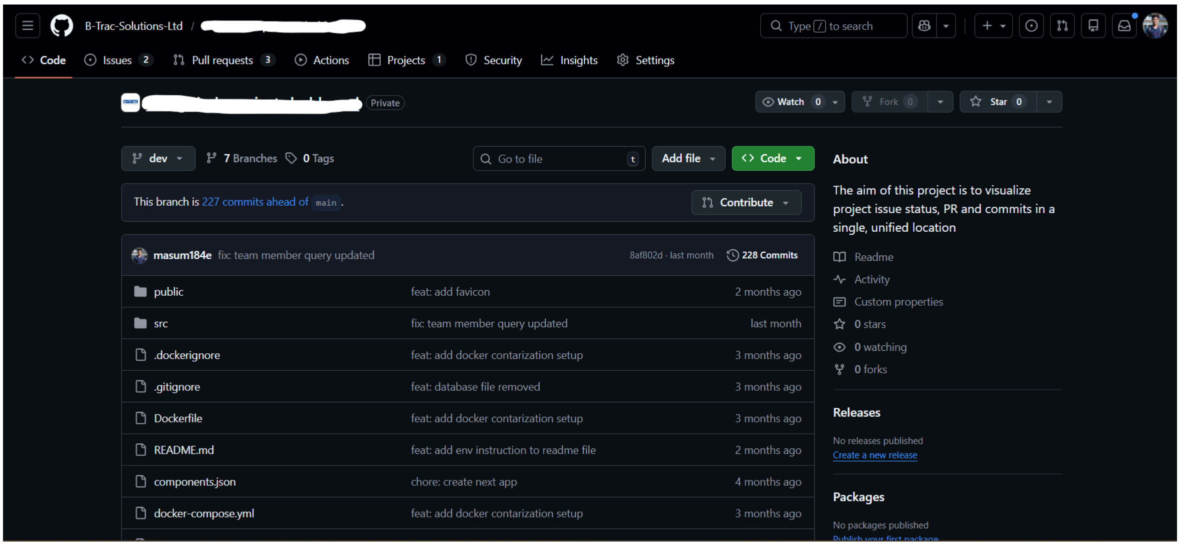


Figure 15: GitHub repository of GitHub dashboard

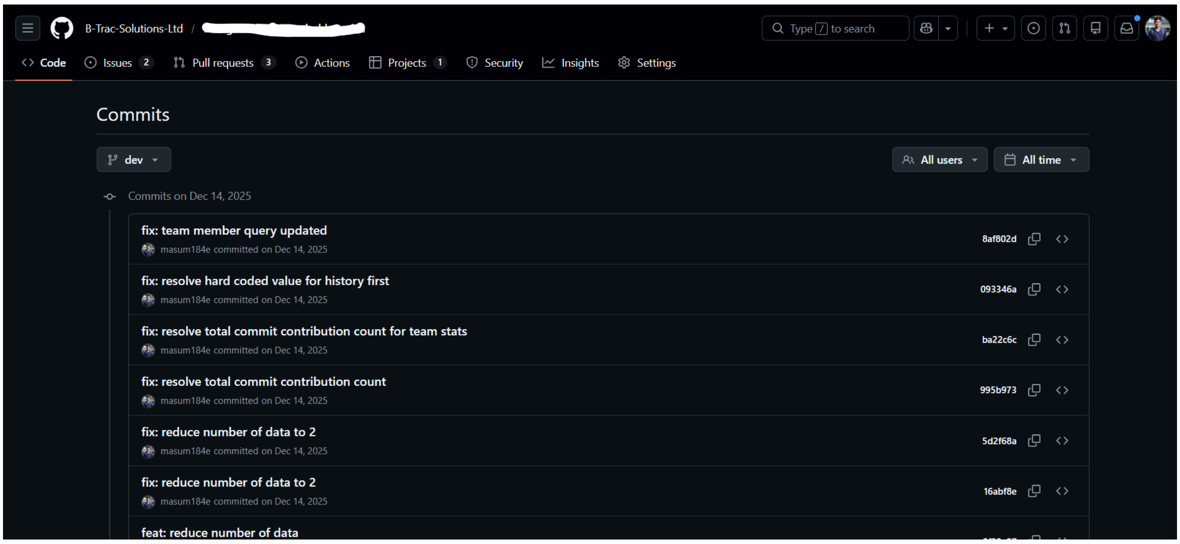


Figure 16: Commit history of GitHub dashboard

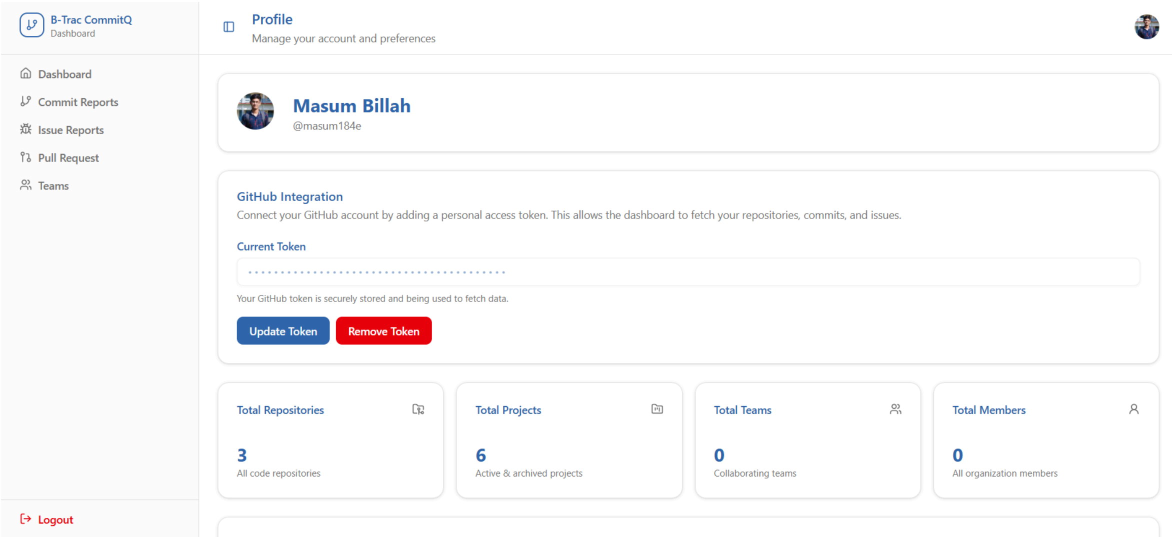


Figure 17: Team leader profile of GitHub dashboard

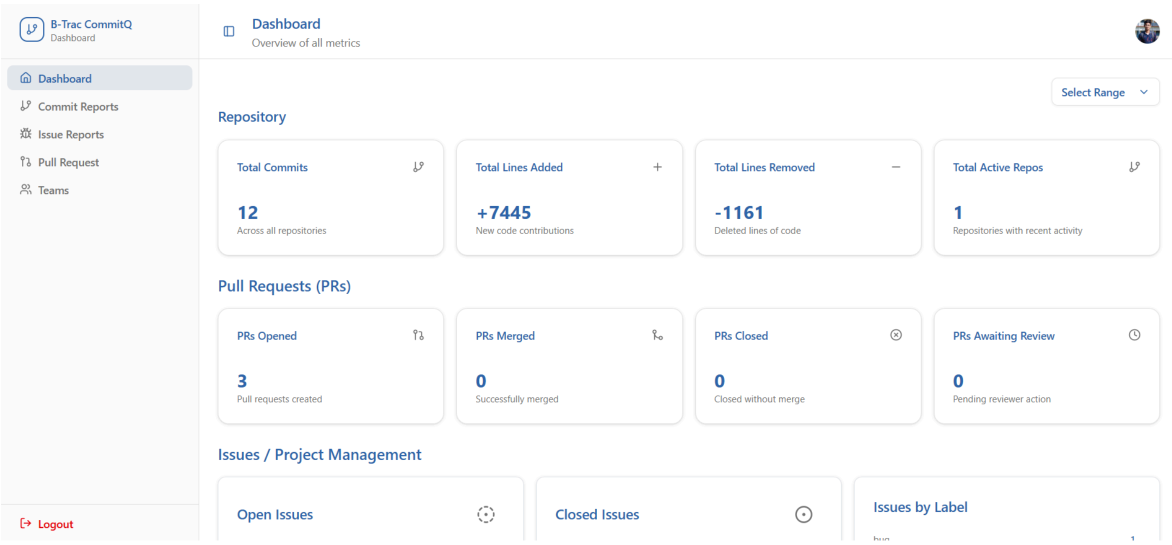


Figure 18: Dashboard of GitHub dashboard

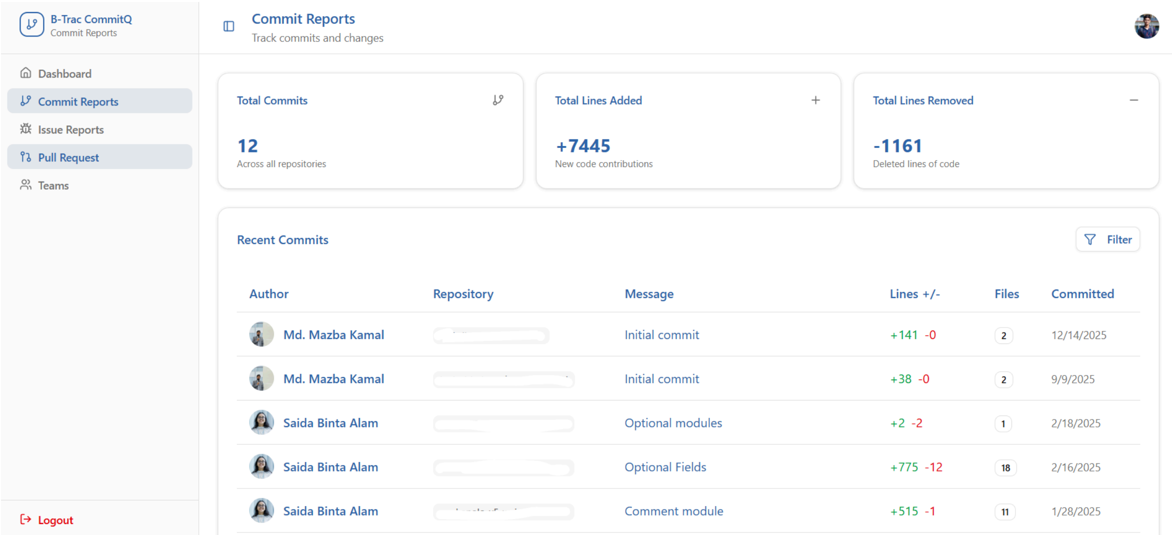


Figure 19: Commit reports of GitHub dashboard

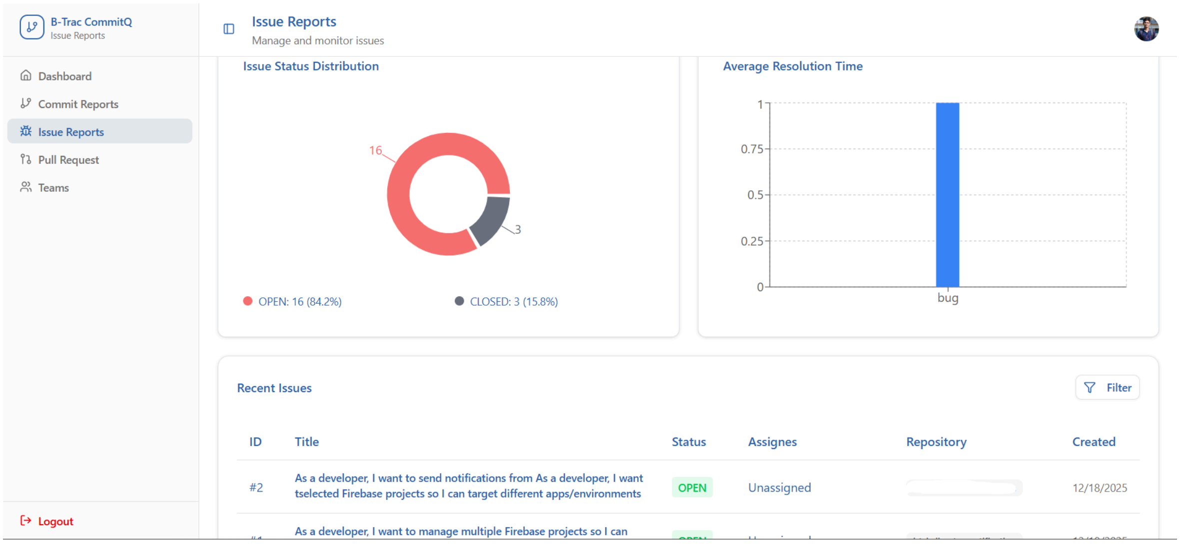


Figure 20: Issue reports of GitHub dashboard

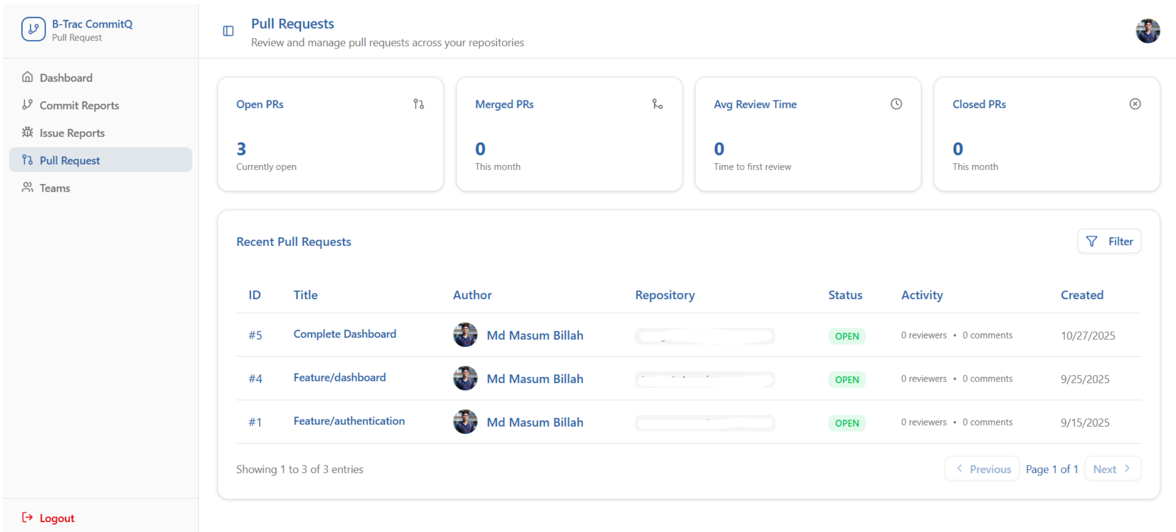


Figure 21: PR reports of GitHub dashboard

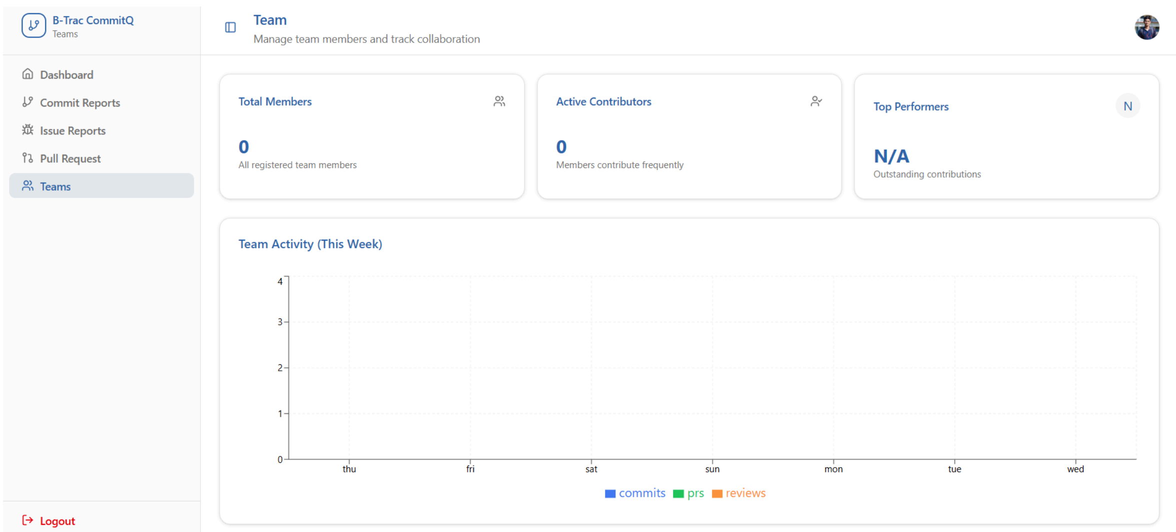


Figure 22: Team analytics of GitHub dashboard

BTSL Disaster Notification

The BTSL Disaster Notification System was an emergency notification application that I developed at B-Trac Solutions Ltd. The aim was to send earthquake alerts to various mobile applications such Workopolo, Nirapath, Carcopolo, and so on when they occurred.

It also retrieves information through the USGS public API and regularly monitors the presence of seismic activities and in case it detects such activity within a certain area, it sends live messages to all clients.

Features

The key features of the BTSL Disaster Notification System include:

- Track earthquake events using the USGS public API based on centralized geographic coordinates
- Periodically poll the public API at 60 second intervals
- Detect earthquake occurrences within specific coordinates
- Send real time notifications to users
- Broadcast notifications to all installed devices
- Support notification delivery to multiple mobile applications simultaneously

Technologies

- TypeScript
- Node.js
- Commander.js
- Firebase Cloud Messaging (FCM)

Learning

This project provided me with the practical experience of design patterns and engineering practices that are solid in a real environment. The fact that the public API was alive posed some issues regarding the continuous polling, event detection and the processing asynchronously.

The use of FCM to do the notifications widened my understanding of real time communication, broadcasting messages, and controlling notifications on multiple applications. This initiative actually increased my skills in designing scalable, reliable, and event driven world backend systems.

Proof of Work

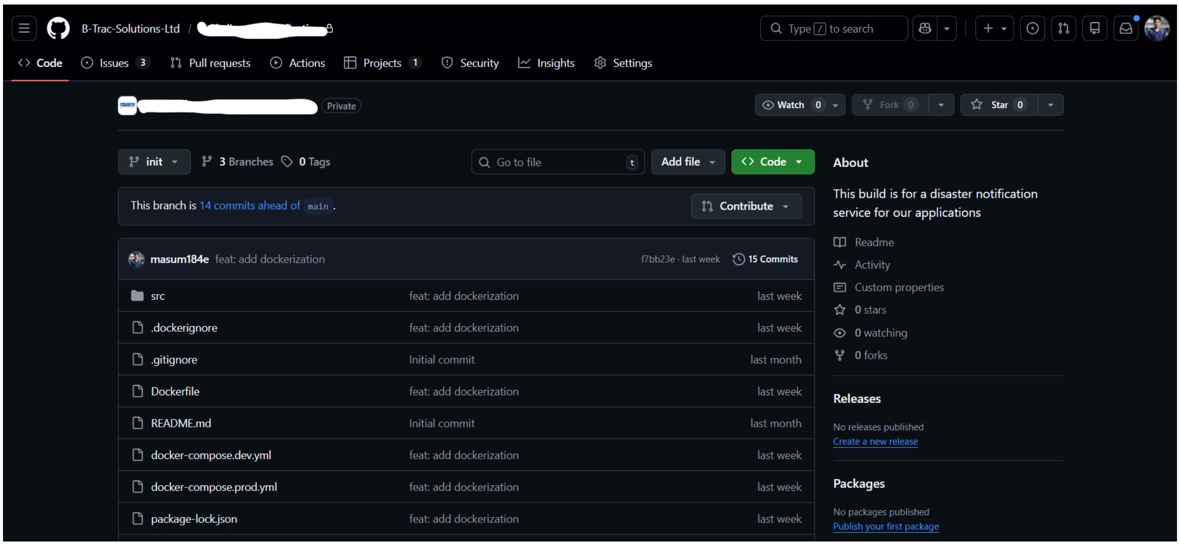


Figure 23: GitHub repository of disaster notification

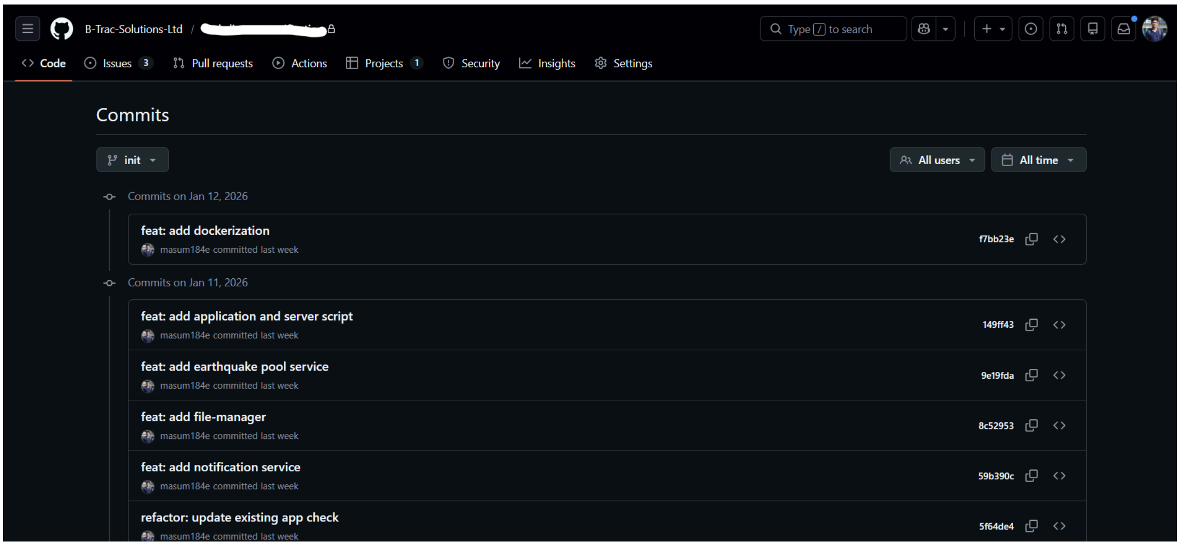


Figure 24: Commit history of disaster notification

My Contributions

Through my internship experience with B-Trac Solutions Ltd, I was able to work on a number of development-related projects. It was an excellent opportunity to apply the theory in my classes in practice, sharpen my software engineering skills and get an idea of how experienced developers do their job.

BTSL GitHub Dashboard

The BTSL GitHub Dashboard, which is a nice internal analytics tool created to increase the productivity of team leaders, was one of the primary issues that I addressed. I have begun by gathering and splitting down the requirements to determine what repository stats and metrics would provide the most value in maintaining a check on team performance and development activity.

Using them, I was able to create a rough sketch of a scalable GraphQL API that can handle giant heaps of GitHub information with ease. The API was created to provide flexibility in queries, reduce unnecessary traffic and remain fast even when retrieving data in numerous repos and contributors. It actually further enhanced my understanding of the backend architecture, data modeling and construction of high speed, high performance APIs.

BTSL Disaster Notification

Another project I did was on the BTSL Disaster Notification System, which is a real time mobile notification service that delivers important earthquake information to users at a time when it is required. The architecture adhered to good practices of software engineering in that way it remained scalable, reliable and simple to maintain.

The task I had was to put together the entire system architecture and write the general logic, but following the general design patterns and industry best practices. I also ensured that I maintained a clean and modular codebase to ensure that any new upgrade and integrations with other mobile apps would not be a problem in the future.

I also connected in real time data downloading of the USGS public API to request valid earthquake data. The solution was fast and sent instant notifications to the customers of any mobile application. The experience of developing it provided me with practical experience in designing real time systems, external API integration, event communication, cross-platform notification strategies.

Learnings

TypeScript

During both the GitHub Dashboard and the Disaster Notification System, I received a lot of exposure to TypeScript since we were required to create these apps practically. My engagement with production grade code helped me better understand the TypeScript type system, both inference and type definition of various types and strict typing, which increases reliability. Because of this, I was able to write more maintainable, safe and scalable code.

Next.JS

During the creation of the GitHub Analytical Dashboard, I immersed myself in Next.js that provided me with a better understanding of contemporary React based stacks. I also practiced shadcn/ui and learned the distinction between client-side and server-side elements, which constitute UI design. NextAuth was also a part that I dealt with particularly in working with the GitHub Personal Access Tokens, how to securely store auth data in the NextAuth state and how changes in state impact the UI and the browser storage. That made me feel more at ease with the auth flows of full stack applications.

GitHub GraphQL API

One of the key lessons was during the work with GraphQL API of GitHub. I compared it with the previous version of REST and understood that GraphQL was much more adaptable and effective for the Dashboard. I was also able to learn how to write optimized queries, and how to write queries to other public GraphQL APIs.

Deployments

I was able to implement applications all the way to production, including pipelines and repairing real life bugs that emerge after the deployment. One of the major hiccups was the handling of the URLs messed up due to a staging server proxy which broke the UI, the API calls and the rendering. This allowed me to stabilize the app again by systematically patching the URLs across the code, and it became a real eye opener to how important the environment conscious development is.

Real Time Communication

When developing Disaster Notification System, I addressed the problem of real time data processing. The application queries the USGS public earthquake alert API on a minute to minute basis in order to capture new events. When one of the earthquakes emerged, the system sent live notifications to any of the mobile applications installed within the organization. This assignment provided me with a good understanding of Firebase Cloud messaging delivering messages to individual devices and to groups.

Design Principles

I had the opportunity to apply the design ideas I gained in my courses to real projects, creating modular architecture, and separating concerns and writing clean code. Those were the habits that enhanced the quality and sustainability of the apps I tried on.

Command Line Tool

At B-Trac Solutions Ltd, I got to know how to create and use command-line tools to support development processes. The experience demonstrated to me how CLIs can increase productivity, automate repetitive tasks and grind out the development.

Quality of Work

I was bumped by GitHub on large data sets with the GraphQL limits when developing the GitHub Analytical Dashboard APIs. I decided to solve this by designing the API to fetch only what is required with intelligent pagination to maintain high performance without exceeding the limits of the API. It provided me with the importance of scalability.

Think Before Code

It was an experience that I should have defined clear requirements before coding so much headache is saved. During the experiments and during the discussions with mentors as the mobbing of the GitHub Analytical Dashboard was progressing the code was reshuffling with new ideas emerging whenever a new idea arose. That demonstrated to me how hard specifications reduced time and rework.

Collaboration

In the case of the notification system, I required an Android test application to test real-time alerts. I collaborated with Saiful Islam Sazib of the Mobile Apps Assistant Manager and we brainstormed over the requirements and developed a lightweight test SDK of Firebase notifications on the fly. It helped me sharpen my communication skills and showed me the importance of teamwork in the process of balancing cross platform work.

Growth

Technical Growth

In my internship, I received the opportunity to use the content of my coursework at the university in actual production ready systems. I studied the latest piles such as TypeScript, Next.js, GraphQL, and the GitHub GraphQL API, and Firebase Cloud Messaging and Firebase Cloud deployment setup automation. Developing the BTSL GitHub Dashboard and the BTSL Disaster Notification System allowed me to do end to end full stack development, API design, real time data pipes, and developing scalable architecture. Another pattern and best practice that I transferred as I learned about software engineering in my classes was design patterns and best practices, and this enabled me to make the link between theory and real life code. In general, I believe that the internship made me a better problem solver, helped me to be more conscious of how code is written, and gave me confidence in approaching more technical problems.

Professional Growth

Other than programming, the internship represented a milestone in my career. I worked in a team atmosphere in which communication and teamwork were significant issues as the code. I also learned how to request feedback, become a member of technical discussions, and how to coordinate my efforts with company tasks and deadlines. Other lessons that the experience offered to me are the expanded knowledge of the workplace ethics, professional duties, and industry standards. I was introduced to the real life experiences of how well organized processes development, deploying cycles, and working in cross functional teams would be useful to my future positions and helped me to leave the reality of the campus life and enter the corporate environment.

Self Assessment

Considering the internship, I believe that it was one of the points of my learning curve, both in personal and professional aspects. Working projects allowed me to see a better image of professional tasks, its workplace relationships, and the industry practices. Never ended learning, conversations with senior colleagues, and solving real life challenges boosted my technical skills and socialization. I am now prepared to approach the future career challenges confidently, with flexibility and feeling of responsibility.

	Outstanding (5)	Very Good (4)	Good (3)	Need Improvements (2)	Poor (1)	Sub-Total
Puntuality Attends office regularly	✓					5
Ability to Solve Problem Ability to identify and resolve the issue	✓					5
Accuracy of Work Correctness in assigned task	✓					5
Creativity Ingenuity shown while solving problems	✓					5
Dedication Completed assigned task within the scheduled time	✓					5
Professionalism Attitude at work environment	✓					5
Teamwork Ability to collaborate with teammember		✓				5
Interpersonal Communication Maintain two way effective	✓					4

communication						
Leadership Take initiative and contribute ideas to improve outcome		✓				4
Growth of New Technology Adaptability to new tools and technologies	✓					5
Total Count (Out of 50)						48

When comparing my performance I concentrated on the abilities that I developed because of my internship. I also tried to be honest and responsible in my tasks with the goal of being a good professional. Although my results were good in most of the areas, I am aware of the areas I can improve on in terms of leadership and more advanced communication. Generally, my self-score of 48 out of 50 is very satisfying as I am proud of the accomplishments and also understand the areas I should continue to work upon.

Time Management

I enhanced my time management by getting both the work that is given to me, learning new technology, and meeting deadlines in a project. Ranking tasks in order of urgency and impact made me deliver on schedule without compromising the quality of the features.

Ability to learn

Since each project was implemented with other tools, frameworks, and API, I was forced to continue learning. My research process included independent studying and consulting mentors where necessary, as well as using new knowledge directly to my work. This increased my learning efficiency in the professional environment.

Accuracy of work

I remained attentive to accuracy through careful requirement reading, testing validation and extensive testing. Determinacy turned into a major issue, particularly when it comes to APIs, real time information, as well as production-grade code; single character flaws can lead to massive problems.

Creativity

Internship has compelled me to be imaginative when dealing with technical problems. During mentor meetings, I found other possible solutions, optimized processes, and pitched ideas. This enhanced my mentality of problem solving and working within the realistic scope.

Dedication and Work Ethic

I demonstrated commitment through being responsible in what I did, and not wavering in my work. I was a self initiator, engaged more in the comprehensiveness of a system, and contributed meaning to the projects I was accomplishing.

Professional Conduct

My experience in a corporate setting taught me that I should be professional in my communication, responsibility, and standard compliance. I also got to know how to respect deadlines, adhere to documentation procedures, and behave professionally during meetings and dialogues.

Team Work and Collaboration

I also had a good experience of teamwork during the internship: the coordination with mentors and other colleagues was important. I took part in serious discussions, exchanged progress, and co-operated to achieve common project objectives.

Adaptation of New Technology

I felt more confident to embrace new technology by addressing the new frameworks and tools that were unknown to me. This experience also helped me to strengthen the value of knowledge about the implementation, study, and use of new technologies in actual projects.

Interpersonal Communication

Frequent communication with mentors and colleagues helped to develop my interpersonal communication. I got to know how to be clear and ask proper questions as well as give structured updates.

Leadership

I provided leadership in the form of identifying issues, offering solutions and coping with work without the directive of my superiors even, as an intern. This was able to make me realize the significance of responsibility, accountability and initiative in the professional environment.

Conclusion

The internship with the B-Trac Solutions Ltd. was an eye-opener that coined my technical capacity and professional psyche. It allowed me to translate the university education into the world of projects, practical modern technologies, and the real part of software development. My future career was well formed on knowledge, skills, and values that I have acquired. The experience did not just enhance my technical competence but also helped me to know clearly what I wanted to achieve as far as my career is concerned and to pursue lifelong learning and development in software engineering.

Citations

1. Billah, Md Masum. “FCM Playground.” *GitHub*, 2026, https://github.com/masum184e/fcm_playground. Accessed 25 01 2026.
2. Billah, Md Masum. “Taskify.” *GitHub*, 2026, https://github.com/masum184e/taskify. Accessed 25 01 2026.
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